

## HTML Forms

<form> is a HTML element to collect input data with containing interactive controls. It provides facilities to input text, number, values, email, password, and control fields such as checkboxes, radio buttons, submit buttons, etc

Forms are generally used when you want to collect data from the user. For example, feedback form, login form, complaint form, inquiry form etc.

### Attributes:

- **action:** The action attribute defines the action to be performed when the form is submitted. Usually, the form data is sent to a file on the server when the user clicks on the submit button.
- **method:** The method attribute specifies the HTTP method to be used when submitting the form data. The form-data can be sent as URL variables (with method="get") or as HTTP post transaction (with method="post").

The default HTTP method when submitting form data is GET.

- **enctype:** The enctype attribute specifies how the form-data should be encoded when submitting it to the server. Note: The enctype attribute can be used only if method="post".

```
<form action="/action.html" method="post" enctype="multipart/form-data">

</form>
```

### GET Method

- Appends the form data to the URL, in name/value pairs
- NEVER use GET to send sensitive data! (the submitted form data is visible in the URL!)
- The length of a URL is limited (2048 characters)
- Useful for form submissions where a user wants to bookmark the result
- GET is good for non-secure data, like query strings in Google

### POST Method:

- Appends the form data inside the body of the HTTP request (the submitted form data is not shown in the URL)
- POST has no size limitations, and can be used to send large amounts of data.
- Form submissions with POST cannot be bookmarked align: It aligns form in left or right or center.

## <input> Tag's "type" attributes

The input element is used to insert various controls in the form, such as text boxes, files, passwords, emails, etc. The input element is applied in the form using <input> tag.

- **<input type="text">**

Defines a one-line text input field.

**Attributes:**

- **maxlength:** Specifies maximum no of characters allowed in textfield.
- **minlength:** Specifies minimum no of characters allowed in textfield.
- **value:** to mention contents of textfield.
- **placeholder:** inform about values to be entered by user
- **name:** defines name of control and works like a variable

- **<input type="password">**

Defines a one-line password input field.

**Attributes:**

- **maxlength:** Specifies maximum no of characters allowed in textfield.
- **minlength:** Specifies minimum no of characters allowed in textfield.
- **value:** to mention contents of textfield.
- **placeholder:** inform about values to be entered by user
- **name:** defines name of control and works like a variable

- **<input type="radio">**

Defines a radio button which allows select one option.

**Attributes:**

- **Checked:** It denotes one of the radio buttons is checked
- **name:** common to all radio buttons of same group.
- **value:** The value attribute defines the unique value associated with each radio button. The value is not shown to the user, but is the value that is sent to the server on "submit" to identify which radio button that was selected.

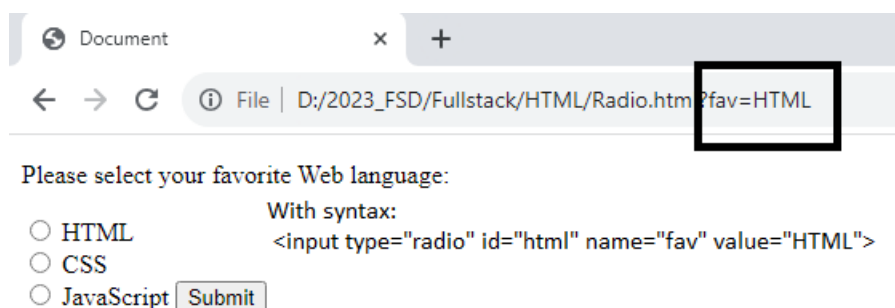
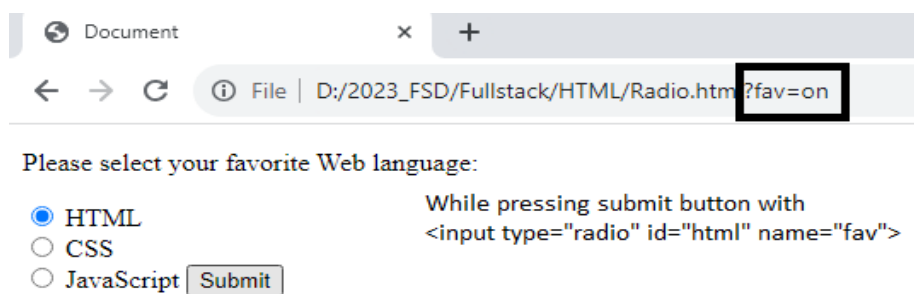
## Example

```
<p>Please select your favourite Web language : </p>
<input type="radio" id="html" name="fav_language" value="HTML">
<label for="html">HTML</label><br>
<input type="radio" id="css" name="fav_language" value="CSS">
<label for="css">CSS</label><br>
<input type="radio" id="javascript" name="fav_language" value="JavaScript">
<label for="javascript">JavaScript</label>
```

### Output:

Please select your favourite Web language:

- ☐ HTML
- ☐ CSS
- ☐ JavaScript



- **<input type="checkbox">**

Defines a checkbox. Checkboxes let a user select ZERO or MORE options of a limited number of choices.

**Attributes:**

- **Checked:** It denotes one of the radio buttons is checked
- **name:** common to all radio buttons of same group.
- **value:** The value attribute defines the unique value associated with each checkbox. The value is not shown to the user, but is the value that is sent to the server on "submit" to identify which checkbox/es that was/were selected.

- **<input type="submit">**

Defines a submit button to submit the form to server.

- **<input type="reset">**

Defines a reset button to reset all values in the form.

- **<input type="button">**

Defines a simple push button, which can be programmed to perform a task on an event.

- **<input type="file">**

Defines to select the file from device storage.

- **<input type="date">**

Defines an input field for selection of date.

- **<input type="email">**

It is used for input fields that should contain an e-mail address.

- **<input type="image">**

Defines an image as a submit button.

- **<input type="number">**

Define a field for entering a number (You can also set restrictions on what numbers are accepted). Use the following attributes to specify restrictions:

- **max:** specifies the maximum value allowed
- **min:** specifies the minimum value allowed
- **step:** specifies the legal number intervals
- **value:** Specifies the default value

- **<input type="search">**

Defines a text field for entering a search string.

**Note:** Remember to set a name for the search field, otherwise nothing will be submitted. The most common name for search inputs is q.

**Example:** `<input type="search" name="q">`

- **<input type="range">**

Defines a range control (like a slider control).

Default range is 0 to 100. However, you can set restrictions on what numbers are accepted with the attributes below.



**Attributes:**

- **max** - specifies the maximum value allowed
- **min** - specifies the minimum value allowed
- **value** - Specifies the default value
- **step** - Specifies the legal number intervals for the value of the input. If you set a step value, the slider will only allow values that are multiples of this step within the defined range.

```
<label for="vol">Volume (between 0 and 50):</label>
<input type="range" id="vol" name="vol" min="0" max="50" step="2">
<input type="submit">
```

## HTML <textarea> tag

The HTML <textarea> tag is used to define a multi-line text input control.

It can hold unlimited number of characters and the texts are displayed in a fixed-width font (usually courier).

**Attributes:**

- **Cols:** Specifies the visible width of a text area.
- **Rows:** Specifies the visible number of lines in a text area.
- **Placeholder:** Specifies a short hint that describes the expected value of a text area.
- **Required:** Specifies that a text area is required/must be filled out.

### HTML <label> tag

The <label> HTML element represents a caption for an item in a user interface.

### HTML <fieldset> and <legend> tags

HTML <fieldset> tag is used to group the logically related fields/labels contained within an HTML form. The use of this tag is optional while creating an HTML form but using <fieldset>, it is easy to understand the purpose of grouped elements of form.

The <legend> tag is used with the <fieldset> element as a first child to define the caption for the grouped related fields.

**<legend>:** The <legend> tag defines a caption for the <fieldset> element.

### HTML <select> tag

<select> tag is used to create a drop-down list with multiple options. The <option> element is nested within <select> tag for defining options in a list.

#### Attributes

- **disabled:** Specifies that an option should be disabled
- **label:** Specifies a shorter label for an option
- **selected:** Specifies that an option should be pre-selected when the page loads.
- **hidden:** The option with the hidden attribute is not visible in the dropdown list, but it remains in the HTML code, and its value can still be accessed programmatically (e.g., through JavaScript).
- **value:** Specifies the value to be sent to a server

**<optgroup> Tag :** element can be used for grouping related options in a list. If you want to send data to server then use <select> tag within <form> element.

The <select> element is used to create a drop-down list. The <select> element is most often used in a form, to collect user input.

#### Attribute:

- **name:** It is needed to reference the form data after the form is submitted (if you omit the name attribute, no data from the drop-down list will be submitted).
- **required:** Specifies that the user is required to select a value before submitting the form.
- **Disabled:** Specifies that an option should be disabled
- **Label:** Specifies a shorter label for an option

**<option> tag :** The <option> tag defines an option in a select list.

### Attributes

- **Disabled:** Specifies that an option should be disabled
- **Label:** Specifies a shorter label for an option
- **Selected:** Specifies that an option should be pre-selected when the page loads
- **Value:** Specifies the value to be sent to a server

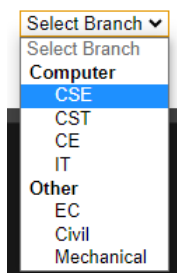
### Optgroup and selected,hidden,disabled example

```
<select name="degree_branch">
  <option value="select" selected disabled> Select Branch</option>
  <optgroup label="Computer">
    <option value="cse"> CSE </option>
    <option value="cst"> CST </option>
    <option value="ce"> CE </option>
    <option value="it"> IT </option>
  </optgroup>
  <optgroup label="Other">
    <option value="ec"> EC </option>
    <option value="civil"> Civil </option>
    <option value="Mech"> Mechanical
</option>
</optgroup>
</select>
```

**For** `<option value="select" selected disabled> Select Branch</option>`

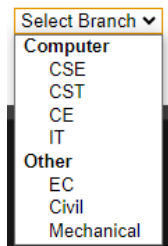
**Selected** attribute keeps the option selected at first time we load the web page.

**Disabled** attribute disables the option. We cannot select the disabled option.



**For** `<option value="select" selected hidden> Select Branch</option>`

**Hidden** attribute hides the option.



**Form tags**

| Tag / Element           | Description                                                  | Syntax / Example                                                                                                                                                                            |
|-------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;form&gt;</b>     | Container used to collect and submit user input to a server. | <code>&lt;form action="/submit" method="post"&gt;<br/>... form elements ...<br/>&lt;/form&gt;</code>                                                                                        |
| <b>&lt;fieldset&gt;</b> | Groups related form elements together inside a box.          | <code>&lt;fieldset&gt;<br/><br/>&lt;/fieldset&gt;</code>                                                                                                                                    |
| <b>&lt;legend&gt;</b>   | Caption/title for a <fieldset>.                              | <code>&lt;fieldset&gt;<br/>  &lt;legend&gt;Login Details&lt;/legend&gt;<br/>&lt;/fieldset&gt;</code>                                                                                        |
| <b>&lt;input&gt;</b>    | General input field (text, password, number, etc.).          | <code>&lt;input type="text" &gt;<br/><br/>&lt;input type="email" &gt;</code>                                                                                                                |
| <b>&lt;textarea&gt;</b> | Multi-line text input field.                                 | <code>&lt;textarea rows="20" cols="30"&gt;Enter your message&lt;/textarea&gt;</code>                                                                                                        |
| <b>&lt;button&gt;</b>   | Clickable button (can be submit, reset).                     | <code>&lt;button type="submit"&gt;Submit&lt;/button&gt;</code>                                                                                                                              |
| <b>&lt;label&gt;</b>    | Defines a label for form elements; improves accessibility.   | <code>&lt;label for="uname"&gt;Username:&lt;/label&gt; &lt;input type="text"&gt;</code>                                                                                                     |
| <b>&lt;datalist&gt;</b> | Provides a list of predefined suggestions for an <input>.    | <code>&lt;input list="browsers" name="browser"&gt;<br/>&lt;datalist id="browsers"&gt;<br/>  &lt;option value="Chrome"&gt;<br/>  &lt;option value="Firefox"&gt;<br/>&lt;/datalist&gt;</code> |
| <b>&lt;select&gt;</b>   | Creates a drop-down list.                                    | <code>&lt;select name="cars"&gt;<br/>  &lt;option value="volvo"&gt;Volvo&lt;/option&gt;<br/>  &lt;option value="bmw"&gt;BMW&lt;/option&gt;<br/>&lt;/select&gt;</code>                       |
| <b>&lt;option&gt;</b>   | Defines an item inside a <select> or <datalist>.             | <code>&lt;select&gt;<br/>  &lt;option value="html"&gt;HTML&lt;/option&gt;<br/>  &lt;option value="css" selected&gt;CSS&lt;/option&gt;<br/>&lt;/select&gt;</code>                            |



## Forms, Graphics, and Interactive Elements

|                         |                                                     |                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>&lt;optgroup&gt;</b> | Groups related <option> elements inside a <select>. | <pre> &lt;select&gt;   &lt;optgroup label="Frontend"&gt;     &lt;option&gt;HTML&lt;/option&gt;     &lt;option&gt;CSS&lt;/option&gt;   &lt;/optgroup&gt;   &lt;optgroup label="Backend"&gt;     &lt;option&gt;PHP&lt;/option&gt;     &lt;option&gt;Python&lt;/option&gt;   &lt;/optgroup&gt; &lt;/select&gt; </pre> |
|-------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Form Attributes

| Attribute   | Description                                                                                                                                                                    | Default Value                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| action      | URL where form data will be sent after submission.                                                                                                                             | "" (empty)                                                                                                             |
| enctype     | The enctype attribute specifies how the form-data should be encoded when submitting it to the server.<br><b>Note:</b> The enctype attribute can be used only if method="post". | application/x-www-form-urlencoded<br>Form data is encoded as key-value pairs, joined with & and spaces converted to +. |
| method      | HTTP method (GET or POST) used when submitting.                                                                                                                                | GET                                                                                                                    |
| type        | Defines the type of <input> (text, password, etc.).                                                                                                                            | text                                                                                                                   |
| name        | Name of the field (used as key when submitting data).                                                                                                                          | "" (empty)                                                                                                             |
| id          | Unique identifier for an element                                                                                                                                               | "" (empty)                                                                                                             |
| value       | Default value for an input.                                                                                                                                                    | "" (empty)                                                                                                             |
| placeholder | Hint text shown inside an input before typing.                                                                                                                                 | "" (empty)                                                                                                             |
| required    | Makes the field mandatory before form submission.                                                                                                                              | false                                                                                                                  |
| readonly    | Field is visible but not editable; value is submitted.                                                                                                                         | false                                                                                                                  |
| disabled    | Field is visible but inactive; value is NOT submitted.                                                                                                                         | false                                                                                                                  |
| checked     | Pre-selects a checkbox or radio button.                                                                                                                                        | false                                                                                                                  |
| maxlength   | Maximum number of characters allowed.                                                                                                                                          | Unlimited                                                                                                              |
| minlength   | Minimum number of characters required.                                                                                                                                         | 0                                                                                                                      |
| step        | Defines intervals for numeric/date inputs (e.g., step='2').                                                                                                                    | 1 (depends on input type)                                                                                              |
| min         | Minimum allowed value for number/date inputs.                                                                                                                                  | No restriction                                                                                                         |
| max         | Maximum allowed value for number/date inputs.                                                                                                                                  | No restriction                                                                                                         |
| multiple    | Allows selecting multiple values (e.g., multiple files or emails).                                                                                                             | false                                                                                                                  |
| accept      | Specifies file types accepted in a file upload input.                                                                                                                          | All file types                                                                                                         |

## Forms, Graphics, and Interactive Elements

| Attribute    | Description                                                  | Default Value                        |
|--------------|--------------------------------------------------------------|--------------------------------------|
| autocomplete | Enables or disables autocomplete suggestions.                | on                                   |
| autofocus    | Automatically focuses on an input field when the page loads. | false                                |
| rows         | Defines visible rows in a <textarea>.                        | No fixed default (browser-dependent) |
| cols         | Defines visible width (columns) in a <textarea>.             | No fixed default (browser-dependent) |
| selected     | Pre-selects an option in a <select> menu.                    | false                                |
| list         | Connects an <input> with a <datalist> for suggestions.       | None                                 |

## <input> Types

| Input Type      | Description                                             | Syntax / Example                                                                                                            |
|-----------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>text</b>     | Single-line text field.                                 | <input type="text" name="username" id="uname" placeholder="Enter username" required maxlength="20">                         |
| <b>password</b> | Hides typed characters (for passwords).                 | <input type="password" name="pwd" id="pwd" placeholder="Enter password" required minlength="6">                             |
| <b>email</b>    | Input for email addresses (validates format).           | <input type="email" name="email" id="email" placeholder="name@example.com" required>                                        |
| <b>number</b>   | Numeric input with min/max/step controls.               | <input type="number" name="age" id="age" min="1" max="100" step="1" value="18">                                             |
| <b>tel</b>      | Telephone number input (may show number pad on mobile). | <input type="tel" name="phone" id="phone" placeholder="1234567890">                                                         |
| <b>checkbox</b> | Allows selecting one or more options.                   | <input type="checkbox" name="hobby" value="music" checked> Music <input type="checkbox" name="hobby" value="sports"> Sports |
| <b>radio</b>    | Allows selecting one option from a group.               | <input type="radio" name="gender" value="male" checked> Male <input type="radio" name="gender" value="female"> Female       |
| <b>submit</b>   | Button to submit the form.                              | <input type="submit" value="Submit Form">                                                                                   |
| <b>reset</b>    | Button to reset all form fields.                        | <input type="reset" value="Reset Form">                                                                                     |
| <b>button</b>   | General button (custom)                                 | <input type="button" value="Click Me">                                                                                      |

|               |                                                  |                                                                                                                 |
|---------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|               | actions via JavaScript).                         |                                                                                                                 |
| <b>file</b>   | Allows file selection/upload.                    | <code>&lt;input type="file" name="resume" id="resume" accept=".pdf,.doc" multiple&gt;</code>                    |
| <b>hidden</b> | Hidden input (not shown to user, but submitted). | <code>&lt;input type="hidden" name="userid" value="12345"&gt;</code>                                            |
| <b>date</b>   | Date picker (calendar).                          | <code>&lt;input type="date" name="dob" id="dob" min="2000-01-01" max="2025-12-31" value="2020-05-15"&gt;</code> |
| <b>range</b>  | Slider control for selecting a numeric value.    | <code>&lt;input type="range" name="vol" id="vol" min="0" max="50" step="5" value="25"&gt;</code>                |
| <b>color</b>  | Color picker input.                              | <code>&lt;input type="color" name="favcolor" id="favcolor" value="#ff0000"&gt;</code>                           |

## readonly, disabled and hidden

### 1. readonly

- **Behavior:** The field is visible and its value can be **submitted** with the form, but the user **cannot edit** it.
- **Use case:** Showing a value that should not be changed but still sent to the server.

### 2. disabled

- **Behavior:** The field is visible but **grayed out, cannot be edited**, and is **not submitted** with the form.
- **Use case:** Temporarily disabling a field based on a condition.

### 3. hidden

- **Behavior:** The field is **not visible at all** on the page, but its value **is submitted** with the form.
- **Use case:** Passing extra data to the server without showing it to the user (like user ID).

### Example

```
<h2>Form Example: readonly, disabled, hidden</h2>
<form action="/submit" method="POST">
  <!-- Readonly field -->
  <label>Username (readonly):</label>
  <input type="text" name="username" value="LJU" readonly>
  <br><br>
  <!-- Disabled field -->
  <label>Promo Code (disabled):</label>
  <input type="text" name="promo" value="SUMMER50" disabled>
```

```

<br><br>
<!-- Hidden field -->
<input type="hidden" name="userId" value="12345">
<!-- Editable field -->
<label>Email:</label>
<input type="email" name="email" placeholder="Enter your email" required>
<br><br>
<button type="submit">Submit</button>
</form>

```

**Form Example: readonly, disabled, hidden**

Username (readonly):

Promo Code (disabled):

Email:

**<input> tag's Attributes**

- **The nameAttribute**

The name attribute specifies the name of an <input> element. The name attribute is used to reference elements in a JavaScript, or to reference form data after a form is submitted. Note: Only form elements with a name attribute will have their values passed when submitting a form.

- **The value Attribute**

The input value attribute specifies an initial value for an input field

- **The readonly Attribute**

The input readonly attribute specifies that an input field is read-only. A read-only input field cannot be modified (however, a user can tab to it, highlight it, and copy the text from it). The value of a read-only input field will be sent when submitting the form!

- **The disabled Attribute**

The input disabled attribute specifies that an input field should be disabled. A disabled input field is unusable and un-clickable. The value of a disabled input field will not be sent when submitting the form!

- **The size Attribute**

The input size attribute specifies the visible width, in characters, of an input field. The default value for size is 20.

- **The maxlength Attribute**

The input maxlength attribute specifies the maximum number of characters allowed in an input field.

**Note:** When a maxlength is set, the input field will not accept more than the specified number of characters.

- **The min and max Attributes**

The input min and max attributes specify the minimum and maximum values for an input field. The min and max attributes work with the following input types: number, range, date.

- **The multiple Attribute**

The input multiple attribute specifies that the user is allowed to enter more than one value in an input field. The multiple attribute works with the following input types: email, and file.

- **The pattern Attribute**

The input pattern attribute specifies a regular expression that the input field's value is checked against, when the form is submitted. The pattern attribute works with the following input types: text, date, search, url, tel, email, and password.

- **The placeholder Attribute**

The input placeholder attribute specifies a short hint that describes the expected value of an input field (a sample value or a short description of the expected format). The short hint is displayed in the input field before the user enters a value.

- **The required Attribute**

The input required attribute specifies that an input field must be filled out before submitting the form. The required attribute works with the following input types: text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.

- **The step Attribute**

The input step attribute specifies the legal number intervals for an input field. Example: if step="3", legal numbers could be -3, 0, 3, 6, etc.

**Tip:** This attribute can be used together with the max and min attributes to create a range of legal values. The step attribute works with the following input types: number, range, date.

- **The autofocus Attribute**

The input autofocus attribute specifies that an input field should automatically get focus when the page loads. The height and width Attributes The input height and width attributes specify the height and width of an <input type="image"> element.

- **The autocomplete Attribute**

The input autocomplete attribute specifies whether a form or an input field should have autocomplete on or off. Autocomplete allows the browser to predict the value. When a user starts to type in a field, the browser should display options to fill in the field, based on earlier typed values. The autocomplete attribute works with <form> and the following <input> types: text, search, url, tel, email, password, datepickers, range, and color.

**\*\*Add on Type for Mobile Number** `<input type="tel">` :Defines a field for entering a telephone number.

**Note:** Browsers that do not support "tel" fall back to being a standard "text" input.

`<label for="phone">Enter your phone number:</label>`

`<input type="tel" id="phone" name="phone" pattern="[0-9]{3}-[0-9]{2}-[0-9]{3}">`

## Example

```
<h2>Student Registration Form</h2>
<form action="submit.html" method="get">
  <!-- Personal Information -->
  <fieldset>
    <legend>Information</legend>

    <label for="uname">Username:</label>
    <input type="text" name="username" id="uname" autofocus
autocomplete="on" placeholder="Enter username" required
maxlength="20"><br><br>

    <label for="pwd">Password:</label>
    <input type="password" name="pwd" id="pwd" placeholder="Enter password"
required minlength="6"><br><br>

    <label for="email">Email:</label>
    <input type="email" name="email" id="email"
placeholder="name@example.com" required><br><br>

    <label for="phone">Phone:</label>
    <input type="tel" name="phone" id="phone"
placeholder="1234567890"><br><br>

    <p>Gender:</p>
    <input type="radio" name="gender" value="male" checked> Male
    <input type="radio" name="gender" value="female"> Female <br><br>

    <p>Hobbies:</p>
    <input type="checkbox" name="hobby" value="music" checked> Music
    <input type="checkbox" name="hobby" value="sports"> Sports <br><br>
  </fieldset>

  <!-- Preferences -->
```

**<fieldset>****<legend>**Additional Information**</legend>****<label for="dob">**Date of Birth:**</label>****<input type="date" name="dob" id="dob" min="2025-09-07" max="2025-12-31">****<br><br>****<label for="age">**Age:**</label>****<input type="number" name="age" id="age" min="1" max="100" step="1" value="18">****<br><br>****<label for="favcolor">**Favorite Color:**</label>****<input type="color" name="favcolor" id="favcolor" value="#ff0000">****<br><br>****<label for="browser">**Preferred Browser:**</label>****<input list="browsers" name="browser" id="browser">****<datalist id="browsers">****<option value="Chrome">****<option value="Firefox">****<option value="Edge">****</datalist>****<br><br>****<label for="degree">**Select Branch:**</label>****<select name="degree\_branch" id="degree">****<option value="select" selected disabled>**Select Branch**</option>****<optgroup label="Computer">****<option value="cse">**CSE**</option>****<option value="cst">**CST**</option>****<option value="ce">**CE**</option>****<option value="it">**IT**</option>****</optgroup>****<optgroup label="Other">****<option value="ec">**EC**</option>****<option value="civil">**Civil**</option>****<option value="mech">**Mechanical**</option>****</optgroup>****</select>****<br><br>****</fieldset>****<fieldset>****<legend>**Feedback**</legend>****<label for="msg">**Your Message:**</label>****<br>****<textarea name="message" id="msg" rows="5" cols="30">**Enter your message**</textarea>****<br><br>**

```

<label for="range">Satisfaction Level:</label>
<input type="range" name="satisfaction" id="range" min="0" max="50" step="5"
value="25"><br><br>
<!-- Hidden value -->
<input type="hidden" name="userid" value="12345">
</fieldset>

<!-- Form Buttons -->
<fieldset>
  <legend>Submit Section</legend>
  <input type="submit" value="Submit Form">
  <input type="reset" value="Reset Form">
  <input type="button" value="Click Me">
  <button type="submit">Submit (Button Tag)</button>
</fieldset>
</form>

```

## Student Registration Form

|                                                                                                                                                                                 |                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>Information</b>                                                                                                                                                              |                                            |
| Username:                                                                                                                                                                       | <input type="text" value="dfg"/>           |
| Password:                                                                                                                                                                       | <input type="password" value="....."/>     |
| Email:                                                                                                                                                                          | <input type="text" value="fh@dfg.gfjh"/>   |
| Phone:                                                                                                                                                                          | <input type="text" value="1234567890"/>    |
| Gender:                                                                                                                                                                         |                                            |
| <input checked="" type="radio"/> Male <input type="radio"/> Female                                                                                                              |                                            |
| Hobbies:                                                                                                                                                                        |                                            |
| <input checked="" type="checkbox"/> Music <input type="checkbox"/> Sports                                                                                                       |                                            |
| <b>Additional Information</b>                                                                                                                                                   |                                            |
| Date of Birth:                                                                                                                                                                  | <input type="text" value="09/16/2025"/>    |
| Age:                                                                                                                                                                            | <input type="text" value="18"/>            |
| Favorite Color:                                                                                                                                                                 | <input type="color" value="#ff0000"/>      |
| Preferred Browser:                                                                                                                                                              | <input type="text" value="Firefox"/>       |
| Select Branch:                                                                                                                                                                  | <input type="text" value="Select Branch"/> |
| <b>Feedback</b>                                                                                                                                                                 |                                            |
| Your Message:                                                                                                                                                                   |                                            |
| <input type="text" value="dgfsdgsdfg"/>                                                                                                                                         |                                            |
| Satisfaction Level: <input type="range" value="25"/>                                                                                                                            |                                            |
| <b>Submit Section</b>                                                                                                                                                           |                                            |
| <input type="button" value="Submit Form"/> <input type="button" value="Reset Form"/> <input type="button" value="Click Me"/> <input type="button" value="Submit (Button Tag)"/> |                                            |

**autofocus and autocomplete='on' added as attributes**



## Student Registration Form

Information

Username:

Password:

Email:

Phone:

Gender: ☒ Male ☐ Female

Saved info

dfg Last used

fh@dfg.gfjh · 1234567890 · Firefox

dfs

fh@dfg.gfjh · 1234567890 · Firefox

dfg

ghfmj

## Example

Design an HTML form that allows users to **upload one or more files** to a server. The form should:

1. Include a file input field where the user can browse and select files.
2. Restrict the file types to only **PDF** (.pdf) and **Word Document** (.doc) formats.
3. Allow users to select **multiple files at once**.
4. Group the file input inside a **fieldset** for better structure.
5. Provide two buttons:
  - **Submit** → to send the selected file(s) to the server.
  - **Reset** → to clear the form and allow new input.
6. Use the **POST** method with `enctype="multipart/form-data"` so the files are correctly transferred in the request body.

```
<h1>File Upload</h1>
<form action="submit.html" method="post" enctype="multipart/form-data">
  <!--Upload File -->
  <fieldset>
    <label for="resume">Upload Resume:</label>
    <input type="file" name="resume" id="resume" accept=".pdf,.doc" multiple><br><br>
  </fieldset>
  <br>
  <input type="submit"/>
  <input type="reset"/>
</form>
```

## File Upload

Upload Resume:  No file chosen

- The **accept attribute** is a **hint to the browser** about what file types should be selectable.
- `<input type="file" accept=".pdf,.doc">` This tells the browser's file picker: *only show PDF and Word files as selectable*.
- The file dialog (file chooser window) will **filter** and only show .pdf and .doc files by default.
- But in many browsers, the user can still change the filter (e.g., "All Files .") and pick any file type (like .png, .exe).
- So accept is **not strict validation**. It's a UI convenience.

### Example:

**Write an HTML form that contains a date picker input. The date picker should only allow users to select dates between January 1, 2020 and December 31, 2022. Ensure that the date field is mandatory.**

```
<html> <body>

<form action="#" method="get">

  <label for="date"> Choose a date (between 01-01-2020 and 31-12-2022) </label>

  <input type="date" name="date" min="2020-01-01" max="2022-12-31"
    required />

  <button type="submit">Submit</button>

</form>

</body></html>
```

### Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0"><head>
  <title>HTML Form</title>
</head>
<body>
<form method="get" action="demo.html">

<fieldset><legend>Signup</legend>
<table>
<tr>
<td>User name </td>
<td><input type="text" maxlength="10" name="username" placeholder="EnterName"/></td>
<!-- autofocus autocomplete="autocomplete" -->
</tr>
<tr>
```

```

<td>Password </td>
<td><input type="password" maxlength="8" placeholder="Enter Password"></td>
</tr>
<tr>
<td>Birth Date </td>
<td><input type="date"></td>
<!-- min="2023-10-01" always in yyyy-mm-dd formate -->
</tr>
<tr>
<td>Contact Number </td>
<td><input type="number"></td>
<!--To put control on number <td><input type="tel" name="tel"/></td> -->
</tr>
<tr>
<td>Email ID </td>
<td><input type="email"></td>
<!-- readonly placeholder="this is readonly field" -->
</tr>
<tr>
<td>Gender </td>
<td>
<input type="radio" name="x" checked/>Male
<input type="radio" name="x"/>Female
<input type="radio" name="x"/>Others
</td>
</tr>
<tr>
<td>Semester</td>
<td><input type="number" max="6" min="1"></td>
</tr>
<tr>
<td>Age:</td>
<td> <input type="range" min="20" max="70" step="1"></td>
</tr>
<!-- To display selected Age in box after using oninput attribute(*Reference)
<tr>
<td> <input type="range" min="20" max="70" step="1" value=""
oninput="num.value = this.value"></td> -->
</tr>
<tr>
<td>Your Age</td>
<td><input type="text" id="num" disabled /></td>
</tr> -->
<tr>

```

```

<td>Location </td>
<td>
<select>
<option selected disabled hidden> - - Select City - - </option>
<optgroup label="GUJARAT">
<option>Ahmedabad</option>
<option>Gandhinagar</option>
<option>Surat</option>
</optgroup>

<optgroup label="MAHARASTRA">
<option>Mumbai</option>
<option>Pune</option>
<option>Kolhapur</option>
</optgroup>
</select>
</td>
</tr>
<tr>
<td>Area Of Intrest </td>
<td>
<input type="checkbox" />DS
<input type="checkbox" />DBMS
<input type="checkbox" checked/>Full Stack
</td>
</tr>
<tr>
<td>Address </td>
<td> <textarea rows="6" cols="20">This has to erase manually</textarea> </td>
</tr>
<tr>
<td>Upload CV </td>
<td><Input type="file" multiple/></td>
</tr>
<tr>
<td><Input type="submit" value="Click me"/></td>
<td><Input type="reset" value="Reset"/></td>
</tr>
<tr>
<td><input type="image" src="2.png" alt="Photo" width="100" height="100"></td>
</tr>
</table>
</fieldset>
</form>
</body></html>

```

**Output:**

Signup

User name

Password

Birth Date  

Contact Number

Email ID

Gender ☒ Male ☐ Female ☐ Others

Odd Semester

Age:

Location

Area Of Intrest ☐ DS ☐ DBMS ☒ Full Stack

Address 

This has to erase manually

Upload CV  No file chosen



**Example: Write a Html file for developing form which include**

1. username with maximum limit of 10 character and placeholder of “EnterName”.
2. Rank between 10 to 100.
3. Range from 5 to 10 with initial value of 7.
4. One search field
5. Submit and reset buttons.

User name

Rank

Rating

Search

```
<html>
<head> <title> HTML Form </title> </head>
<body>
  <form>
    <table>
      <tr>
        <td>User name </td>
        <td><input type="text" maxlength="10" placeholder="EnterName"/></td>
      </tr>
      <tr>
        <td>Rank </td>
        <td><input type="number" min="10" max="100"/></td>
      </tr>
      <tr>
        <td>Rating </td>
        <td><input type="range" max="10" min="5" value="7"/></td>
      </tr>
    </table>
  </form>
  <div>
    <input type="text" value="Search the site..."/>
    <input type="button" value="Submit"/>
    <input type="button" value="Reset"/>
  </div>
</body>
</html>
```

```
</tr>
<tr>
  <td>Search </td>
  <td><input type="search" placeholder="Search the site..." /></td>
</tr>
<tr>
  <td><input type="submit" /></td>
  <td><input type="reset" /></td>
</tr>
</table>
</form>
</body>
</html>
```

## HTML SVG

- SVG defines vector-based graphics in XML format.
- SVG stands for Scalable Vector Graphics
- SVG is used to define graphics for the Web
- SVG is a W3C recommendation
- The HTML `<svg>` element is a container for SVG graphics.
- SVG has several methods for drawing paths, boxes, circles, text, and graphic images.
- It does not lose quality when resized to max.
- Unlike raster image (Ex .jpg, .gif, .png, etc.), a vector image can be scaled up or down to any extent without losing the image quality.

### **SVG Advantages**

Advantages of using SVG over other image formats (like JPEG and GIF) are:

- SVG images can be created and edited with any text editor
- SVG images can be searched, indexed, scripted, and compressed
- SVG images are scalable
- SVG images can be printed with high quality at any resolution
- SVG images are zoomable
- SVG graphics do NOT lose any quality if they are zoomed or resized
- SVG is an open standard
- SVG files are pure XML

### **`<svg>` tag attributes:**

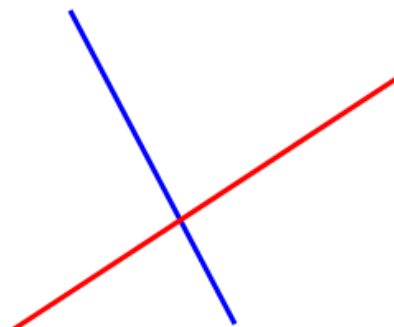
|        |                                                                                            |
|--------|--------------------------------------------------------------------------------------------|
| height | The height attribute is used to specify the vertical length of the SVG element in pixels.  |
| width  | The width attribute is used to specify the horizontal length of the SVG element in pixels. |

**SVG <Line>**

|              |                                             |
|--------------|---------------------------------------------|
| X1           | defines the start of the line on the x-axis |
| Y1           | defines the start of the line on the y-axis |
| X2           | defines the end of the line on the x-axis   |
| Y2           | defines the end of the line on the y-axis   |
| Stroke       | defines the colour of the line              |
| Stroke-width | defines the width of the line               |

**Example1:**

```
<svg height="250" width="500" >  
  <line x1="100" y1="10" x2="200" y2="200" stroke="blue" stroke-width="3"></line>  
  <line x1="300" y1="50" x2="40" y2="220" stroke="red" stroke-width="3"/>  
</svg>
```

**Output:****Code explanation:**

- The x1 attribute defines the start of the line on the x-axis
- The y1 attribute defines the start of the line on the y-axis
- The x2 attribute defines the end of the line on the x-axis
- The y2 attribute defines the end of the line on the y-axis
- The stroke-width attribute defines the width of the border of the line
- The stroke attribute defines the color of the border of the line.



The stroke attribute must be set to a color (e.g., black, red, #000000) for the shape's outline to appear. The **default stroke** is not applied automatically, and an unspecified stroke means no visible stroke at all

**Default value is not white. But we need to define a stroke attribute for the shape's outline to appear.**

### SVG Rectangle

The **<rect>** element is used to create a rectangle and variations of a rectangle shape.

#### SVG <rect> tag attributes:

|              |                                                     |
|--------------|-----------------------------------------------------|
| x            | The x coordinate of the rect. (*default value is 0) |
| y            | The y coordinate of the rect. (*default value is 0) |
| width        | It defines width of the rectangle.                  |
| height       | It defines height of the rectangle.                 |
| Stroke-width | defines the width of the border of the rectangle.   |
| Stroke       | defines the colour of the border of the rectangle.  |
| Fill         | defines the fill colour of the rectangle.           |
| rx           | Specifies rounding value of x-axis.                 |
| ry           | Specifies rounding value of y-axis.                 |

#### Example 1:

The <rect> element is used to create a rectangle and variations of a rectangle shape:

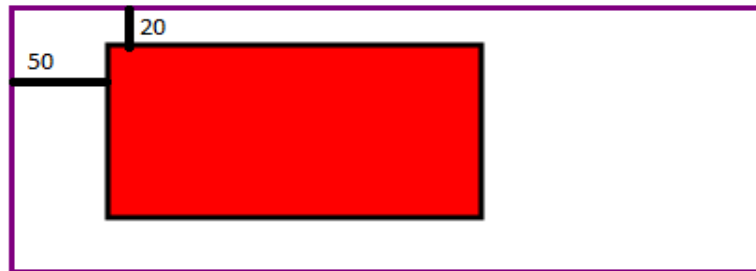
```
<svg width="400" height="120" style="border:3px solid purple">  
  <rect width="300" height="100" stroke="black" stroke-width="3" fill="red" />  
</svg>
```



In above example by default x and y coordinates are (0, 0)

- The width and height attributes of the <rect> element define the height and the width of the rectangle
- The fill attribute defines the fill color of the rectangle.
- The stroke-width attribute defines the width of the border of the rectangle.
- The stroke attribute defines the color of the border of the rectangle.

```
<svg width="400" height="150" style="border:3px solid purple">  
  <rect x="50" y="20" width="200" height="100" stroke="black" stroke-  
width="3" fill="red" />  
</svg>
```



- The x attribute defines the left position of the rectangle (e.g. x="50" places the rectangle 50 px from the left margin)
- The y attribute defines the top position of the rectangle (e.g. y="20" places the rectangle 20 px from the top margin)

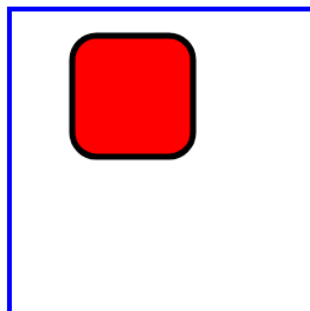
### SVG Rounded Rectangle

The rx and ry attributes of <rect> tag rounds the corner of rectangle.

#### Example 2:

```
<html><body>  
  <svg height="250" width="250" style="border:4px solid blue">  
    <rect x="50" y="20" rx="20" ry="20" width="100" height="100" fill="red"  
stroke="black" stroke-width="5"/>  
  </svg>  
</body>  
</html>
```

#### Output:



## SVG Circle

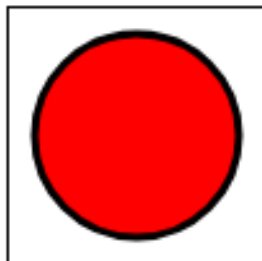
The <circle> element is used to create a circle:

### **SVG <circle> tag attributes:**

|              |                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------|
| cx           | The x coordinate of the center of the circle.                                                               |
| cy           | The y coordinate of the center of the circle. If cx and cy are omitted, the circle's center is set to (0,0) |
| r            | It defines radius of the circle.                                                                            |
| Stroke-width | defines the width of the border of the circle.                                                              |
| Stroke       | defines the colour of the border of the circle.                                                             |
| Fill         | defines the fill colour of the circle.                                                                      |

### **1. Example:**

```
<svg height="100" width="100" style="border:1px solid black">  
  <circle cx="50" cy="50" r="40" stroke="black" stroke-width="3" fill="red" />  
</svg>
```



### **Code explanation:**

- The cx and cy attributes define the x and y coordinates of the center of the circle. If cx and cy are omitted, the circle's center is set to (0,0)
- The r attribute defines the radius of the circle.
- The fill attribute defines the fill color of the circle.
- The stroke-width attribute defines the width of the border of the circle.
- The stroke attribute defines the color of the border of the circle.

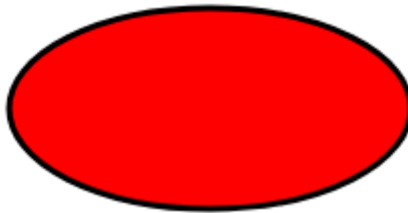
## SVG Ellipse

The <ellipse> element is used to create an ellipse.

An ellipse is closely related to a circle. The difference is that an ellipse has an x and a y radius that differs from each other, while a circle has equal x and y radius:

**Example:**

```
<svg height="140" width="500">  
  <ellipse cx="200" cy="80" rx="100" ry="50" stroke="black" stroke-width="3" fill="red"/>  
</svg>
```



**Code explanation:**

- The cx attribute defines the x coordinate of the center of the ellipse
- The cy attribute defines the y coordinate of the center of the ellipse
- The rx attribute defines the horizontal radius
- The ry attribute defines the vertical radius.
- The fill attribute defines the fill color of the ellipse.
- The stroke-width attribute defines the width of the border of the ellipse.
- The stroke attribute defines the color of the border of the ellipse.

## SVG Polygon

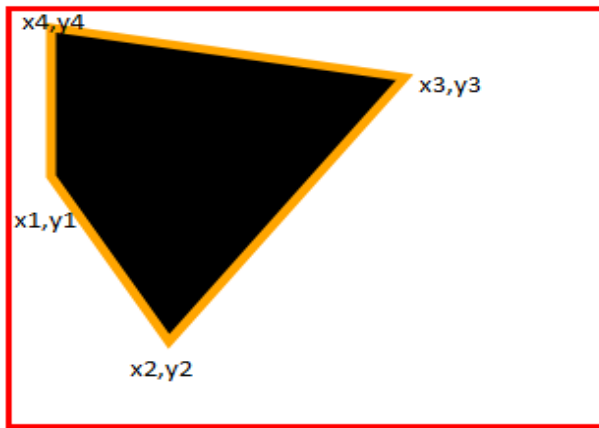
Polygons are made of straight lines, and the shape is "closed" (all the lines connect up). Polygon comes from Greek. "Poly" means "many" and "gon" means "angle".

<polygon> element is used to draw a closed shape consisting of connected straight lines.

The <polygon> element is used to create a graphic that contains at least three sides.

Polygons are made of straight lines, and the shape is "closed" (all the lines connect up).

```
<polygon points="x1,y1,x2,y2,x3,y3,x4,y4....xn,yn" stroke="color" stroke-width="5"  
fill="black" />
```



### Code explanation:

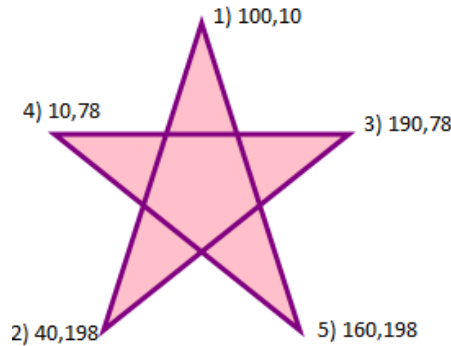
- The points attribute defines the x and y coordinates for each corner of the polygon
- The fill attribute defines the fill color of the polygon.
- The stroke-width attribute defines the width of the border of the polygon.
- The stroke attribute defines the color of the border of the polygon.

### Attributes:

|              |                                                                                                                                                                                                                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| points       | List of points to make up a polygon. "x1,y1,x2,y2,x3,y3,x4,y4....xn,yn"                                                                                                                                                                                                                                                   |
| Stroke       | defines the colour of the border of the circle.                                                                                                                                                                                                                                                                           |
| Stroke-width | defines the width of the border of the circle.                                                                                                                                                                                                                                                                            |
| Fill         | defines the fill colour of the circle. (If not specify default is black)                                                                                                                                                                                                                                                  |
| fill-rule    | <p>The algorithm to use to determine the <i>inside</i> part of a shape. After counting the crossings paths, if the result is zero then the point is outside the path. Otherwise, it is inside.</p> <p>Two possible values – 1) nonzero 2) evenodd</p> <p><b>Default value is “nonzero” (Fill all portion in poly)</b></p> |

### Use the <polygon> element to create a **Star**:

```
<svg height="210" width="500">
  <polygon points="100,10,40,198,190,78 10,78,160,198"
    stroke="purple" stroke-width="3" fill="pink" fill-rule="nonzero" />
</svg>
```



- ✓ The **fill-rule** attribute is a presentation attribute defining the algorithm to use to determine the *inside* part of a shape.
- ✓ After counting the crossings paths, if the result is zero then the point is outside the path. Otherwise, it is inside. **Default value is “nonzero”**

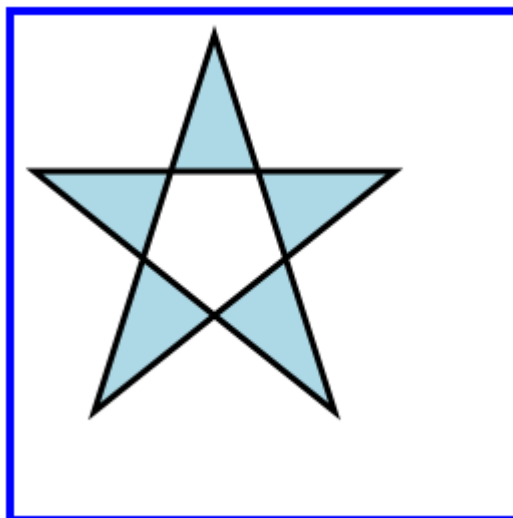
### Example:

```
<html><body>
  <svg height="250" width="250" style="border:4px solid blue">

    <!-- for triangle polygon points="100 10,50 60,150 60" fill="lightblue" stroke="black" stroke-
    width="3" fill-rule="evenodd"/> -->
    <polygon points="100 10,40 198,190 78,10 78,160 198" fill="lightblue" stroke="black"
    stroke-width="3" fill-rule="evenodd"/>

  </svg>
</body></html>
```

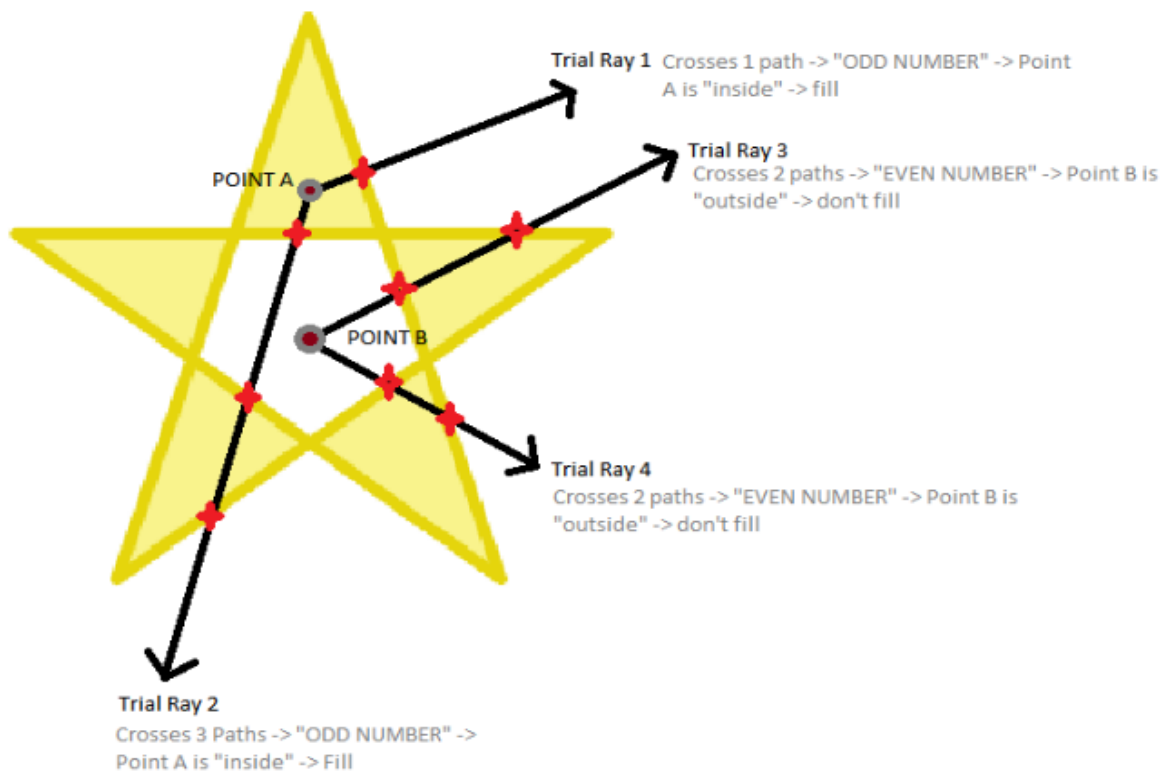
### Output:



## Fill-rule = “evenodd” concept

This is how the mechanism works. Pick a point in the center, draw lines to infinity in any direction - they always cross an even number of path segments - which means that they are “outside” and their areas don’t get filled.

Pick a point in the filled triangles - if you draw lines to infinity in any direction - they always cross an odd number of path segments and thus, they are “inside” and their area should be filled.



### Note:

- ✓ If the **fill** attribute is not specified, then **default value is black**.
- ✓ **stroke** attribute is necessary to define in **line tag**, otherwise **line** will not be visible.
- ✓ In other tags like **rect**, **circle**, **polygon**, **ellipse** etc it will fill the shape with **default color black/mentioned color** without stroke (border of the shape).

SVG shapes like <circle>, <rect>, <line> etc do not render a stroke unless a stroke color is explicitly provided. **There is no default stroke automatically applied unless specified.**

**Without the stroke attribute, the circle won't render any visible outline because no stroke color is applied.** To make the circle visible, you need to explicitly define the stroke color.

**Default value is not white. But we need to define a stroke attribute for the shape's outline to appear.**

## SVG Logo

The <ellipse> element is used to create an ellipse.

An ellipse is closely related to a circle. The difference is that an ellipse has an x and a y radius that differs from each other, while a circle has equal x and y radius:

### **Attributes:**

|    |                                                |
|----|------------------------------------------------|
| cx | The x coordinate of the center of the ellipse. |
| cy | The y coordinate of the center of the ellipse. |
| rx | It defines horizontal radius.                  |
| ry | It defines vertical radius.                    |

```
<svg height="140" width="500">
  <ellipse cx="200" cy="80" rx="100" ry="50" stroke="black" stroke-width="3" fill="pink"/>
  <text fill="#000" font-size="50" x="155" y="100" font-family="verdana">LJU</text>
</svg>
```



### **Example 1:**

```
<html><body>

  <svg width="250" height="250">
    <rect width="100%" height="100%" fill="grey" />
    <circle cx="120" cy="120" r="80" fill="cyan" />
    <text x="70" y="130" font-size="60" fill="white">Svg</text>
  </svg>

</body></html>
```

### **Output:**





**Example 2**

```
<html>

<body>

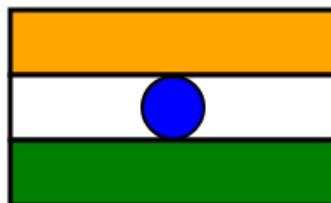
  <svg width="250" height="200">
    <rect width="100%" height="100%" fill="grey" />
    <ellipse cx="120" cy="100" rx="85" ry="55" fill="cyan" />
    <text x="70" y="120" font-size="60" fill="white">Svg</text>
  </svg>

</body>

</html>
```

**Output:****Example 3: Draw Flag**

```
<svg height="250" width="500">
  <rect x='20' y='20' width="200" height="40" fill="orange" stroke="#000" stroke-
width="3"/>
  <rect x='20' y='60' width="200" height="40" fill="#fff" stroke="#000" stroke-
width="3"/>
  <rect x='20' y='100' width="200" height="40" fill="green" stroke="#000" stroke-
width="3"/>
  <circle cx="120" cy="80" r="19" stroke="#000" stroke-width="2" fill="blue"/>
</svg>
```



**Example 4** Draw a star in SVG HTML.

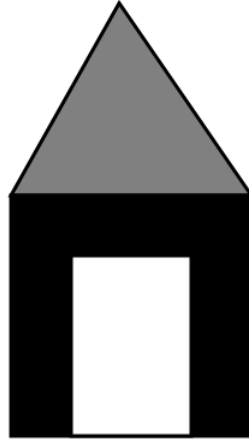
```
<!DOCTYPE html>
<html>
<body>
<svg width="300" height="200">
  <polygon points="100,10 40,198 190,78 10,78 160,198"
    style="fill:orange;stroke:black;stroke-width:5;fill-rule:nonzero;" /> // evenodd can be used in fill-rule
  Sorry, your browser does not support inline SVG.
</svg>
</body>
</html>
```

**Example 1**

```
<svg height="250" width="500">
  <rect x='20' y='20' width="200" height="40" fill="orange" stroke="#000" stroke-
width="3"/>
  <rect x='20' y='60' width="200" height="40" fill="#fff" stroke="#000" stroke-width="3"/>

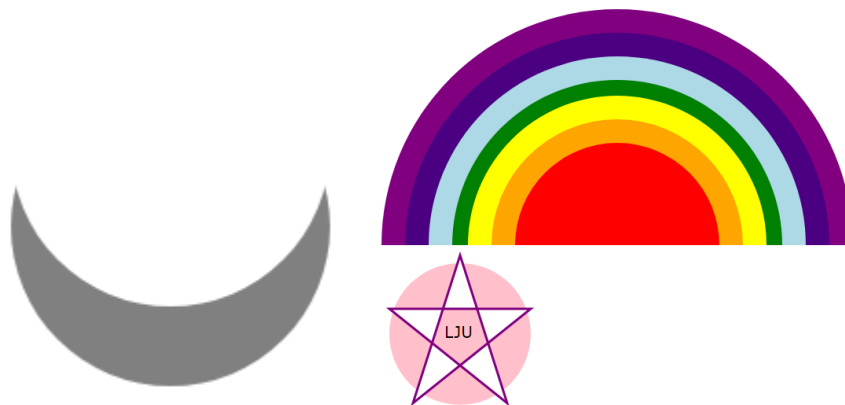
  <rect x='20' y='100' width="200" height="40" fill="green" stroke="#000" stroke-
width="3"/>
  <circle cx="120" cy="80" r="19" stroke="#000" stroke-width="2" fill="blue"/>
</svg>
```

## Example 2



```
<svg height="450" width="500">
  <polygon points="20,180 110,20 220,180" fill="grey" stroke="black" stroke-width="3"
/>
  <rect x='20' y='180' width="200" height="200" fill="black" stroke="#000" stroke-
width="3"/>
  <rect x='70' y='230' width="100" height="150" fill="white" stroke="#000" stroke-
width="3"/>
</svg>
```

## Example 3



```
<!--moon -->
<svg width="200" height="200">
  <circle cx="120" cy="100" r="80" fill="grey"/>
  <circle cx="120" cy="60" r="80" fill="white"/>
</svg>

<!--rainbow half circle -->
<svg width="600" height="500">
  <circle cx="300" cy="500" r="300" fill="purple"/>
  <circle cx="300" cy="500" r="270" fill="indigo"/>
```

```

<circle cx="300" cy="500" r="240" fill="lightblue"/>
<circle cx="300" cy="500" r="210" fill="green"/>
<circle cx="300" cy="500" r="190" fill="yellow"/>
<circle cx="300" cy="500" r="160" fill="orange"/>
<circle cx="300" cy="500" r="130" fill="red"/>
</svg>

```

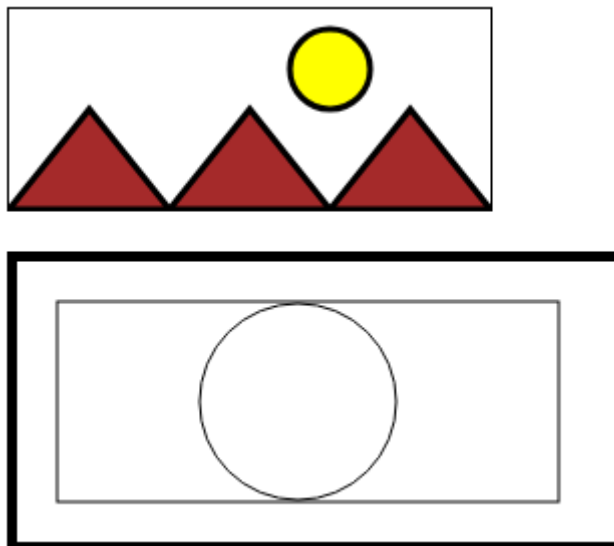
### <!--star circle logo -->

```

<svg height="400" width="500">
  <circle cx="100" cy="110" r="90" fill="pink"></circle>
  <polygon points="100,10,40,198,190,78 10,78,160,198"
    stroke="purple" stroke-width="3" fill="white" fill-rule="evenodd" />
  <text fill="#000" font-size="20" x="80" y="115" font-family="verdana">LJU</text>
</svg>

```

## Example 4



```

<svg height="100" width="240" style="border: 1px solid black;">
  <polygon points="0,100,40,50,80,100,80,100,120,50,160,100,160,100,200,50,240,100"
    fill="brown" stroke="black" stroke-width="3"/>
  <circle cx="160" cy="30" r="20" fill="yellow" stroke="black" stroke-width="3"/>
</svg>
<br>

<svg height="140" width="300" style="border: 5px solid black;">
  <rect x="20" y="20" width="250" height="100" fill="white" stroke="black" stroke-
width=""/>
  <circle cx="140" cy="70" r="49" fill="#fff" stroke="#000"/>
</svg>

```

## HTML Image Maps

With HTML image maps, you can create clickable areas on an image.

The HTML `<map>` tag defines an image map. An image map is an image with clickable areas. The areas are defined with one or more `<area>` tags.

### ❖ `<map>` tag attributes:

**name:** The map tag is mainly associated with the “name” attribute. The name attribute is used to give a name to that particular defined map. This name is then used in the img element so that they both can be linked together.

**Note:** The usemap attribute in `<img>` is associated with the `<map>` element's name attribute, and creates a relationship between the image and the map.

### ❖ `<area>` tag attributes:

|        |                                                                                     |
|--------|-------------------------------------------------------------------------------------|
| shape  | Specifies the shape of the area. It can be a “rect”, a “circle” or a “poly”.        |
| coords | Specifies the coordinates of the area                                               |
| href   | Specifies the hyperlink target for the area                                         |
| target | Specifies where to open the target URL                                              |
| alt    | Specifies an alternate text for the area. Required if the href attribute is present |

### Points for Coords Attribute in `<area>` tag

| For shape | coords                |                                                                                                                                                                                                      |
|-----------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rect      | x1,y1,x2,y2           | Specifies the coordinates of the top-left and bottom-right corner of the rectangle (shape="rect")                                                                                                    |
| circle    | x,y,radius            | Specifies the coordinates of the circle center and the radius (shape="circle")                                                                                                                       |
| poly      | x1,y1,x2,y2,...,xn,yn | Specifies the coordinates of the edges of the polygon.<br>If the first and last coordinate pairs are not the same, the browser will add the last coordinate pair to close the polygon (shape="poly") |

### Example:

```
<html>

<head></head>



<map name ="#workmap">

<area shape="rect" coords="5, 10, 200, 400" href="apple.png" target="_blank"/>

<area shape="rect" coords="210, 10, 400, 400" href="kiwi.png" target="_blank"/>

</map>

</html>
```

**Note:** The usemap attribute in <img> is associated with the <map> element's name attribute, and creates a relationship between the image and the map.

to find coordinates: <https://www.image-map.net/>

### Output:



After clicking on rectangle coordinates 5, 10, 200, 400



After clicking on rectangle coordinates 210, 10, 400, 400

